REMOTE SENS. ENVIRON. 41:197–209 (1992)

Accurate Land Surface Temperature Retrieval from AVHRR Data with Use of an Improved Split Window Algorithm

Yann H. Kerr
LERTS, Toulouse, France

Jean Pierre Lagouarde
INRA Bioclimatologie, Bordeaux, France

Jacques Imbernon
IRAT, Montpellier, France

In order to retrieve accurate values of land surface temperatures from satellite thermal infrared data, it is necessary to perform atmospheric corrections. This paper describes how a differential absorption method, similar to the split window technique used over sea surfaces, can be adapted and used locally over land surfaces. Work previously performed by several authors is first described, allowing one to stress the different issues to be solved, that is, the emissivity and angular effects. A semiempirical algorithm is then described, and the results obtained with it over two test areas are then described and discussed. The proposed method allows to derive land surface temperature with an accuracy better than 1.5 K. A possible algorithm is also suggested.

INTRODUCTION

Excepted solar irradiance components, most of the fluxes at the surface / atmosphere interface can only be parameterized through the use of surface temperature. Land surface temperature can play either a direct role, such as when estimating long-wave fluxes, or indirectly, such as when estimating latent and sensible heat fluxes. Moreover, many other applications rely on the knowledge of land surface temperature [geology, hydrology, vegetation monitoring, global scale models (GCM)]. Consequently, for many studies, it is crucial to have access to reliable estimates of surface temperature over large spatial and temporal scales.

As it is practically impossible to obtain such information from ground-based measurements, the use of satellite measurements in the thermal infrared appears to be very attractive since it can give access to global and uniform estimates of surface temperature. As a matter of fact, satellite thermal infrared sensors measure a radiance that can be translated into top of the atmosphere brightness temperature. If the sensor is designed to work in a part of the spectrum where the atmosphere is almost transparent (10.5–12.5 μm, for example), access to surface temperature seems to be an easy task. It is not generally the case,

Address correspondence to Yann H. Kerr, LERTS, 18 Ave. E. Belin, 31055, Toulouse Cedex, France.
Received 1 March 1992.

0034-4257 / 92 / $5.00
©Elsevier Science Publishing Co. Inc., 1992
655 Avenue of the Americas, New York, NY 10010

due to the fact that the atmosphere, even though almost transparent, still has a nonnegligible effect. Moreover, the surface emissivity is almost always unknown when land surfaces are not black or even grey bodies (i.e., the emissivity is frequency-dependent). To sum up, with satellites we have a means of deriving spatial and temporal values of surface temperature provided that we can perform accurate atmospheric corrections and provided that we can take into account the surface emissivity.

In this article, we are going to see whether it is possible to correct for the atmosphere with the differential absorption method, so as to derive surface brightness temperatures. We are going to study the particular case of the Advanced Very High Resolution Radiometer/2 (AVHRR/2) on board the National Oceanic and Atmospheric Administration (NOAA) polar orbiter series. The study is aimed toward the determination of land surface temperature from satellite data. After some general considerations we will describe the underlying theory and present a semiempirical algorithm together with the results obtained with it in two different environments.

THEORY BACKGROUND

Without going into too much details, we will now give the very basic concepts necessary to define the problem.

Role of the Atmosphere

The energy going through an elementary solid angle per unit time and unit wavelength can be written (Chandrasekhar, 1960) as

$$dI_\lambda / ds = (I_\lambda + B_\lambda)\tau_\lambda , \qquad (1)$$

where I_λ is the intensity of radiation at wavelength λ passing through an absorbing and emitting layer, s the path length, B_λ the blackbody emission of the layer given by the Plank function, and τ_λ the optical depth. After integrating Eq. (1) along the complete path between the surface and the top of the atmosphere, we have

$$\begin{aligned} I_\lambda = {} & \epsilon_\lambda B_\lambda(T_s) \ \exp(-\tau_\lambda^*) \\ & + \int_0^{\tau_\lambda^*} B_\lambda(T(\tau_\lambda)) \ \exp(-\tau_\lambda) \ d\tau_\lambda \qquad (2) \\ & + (1-\epsilon_\lambda)[\int_{\tau_\lambda^*}^{0} B_\lambda(T(\tau_\lambda)) \ \exp(-\tau_\lambda) \ d\tau_\lambda] \ \exp(-\tau_\lambda^*) . \end{aligned}$$

The first term is related to the surface contribution, the second to the atmospheric contribution along the upward path, and the third to the atmospheric contribution along the downward path, reflected by the surface and attenuated along the upward path. Equation (2) indicates that it is necessary to take into account the atmosphere effects and to make measurements in a spectral region where the atmospheric contribution is as small as possible. In most cases the satellite-borne sensors are designed to work in one of the "atmospheric windows" (10.5–12.5 μm in this study). When this condition is met, the first term of Eq. (2) will be least affected, while the relative importance of the second term will be very variable, depending upon meteorological conditions (a thin cirrus will have a significant influence, for instance). The role of third term is related to the surface characteristics: the larger the emissivity, the smaller the contribution.

In this study we neglect scattering in the atmosphere, this effect being small when visibility is higher than 5 km (McClatchy et al., 1971). We also neglect the influences of carbon dioxide (CO_2) and ozone (O_3) as they are much smaller than the effect of water vapor. Finally, we assume that the characteristics of the sensor (normalized response function, calibration) are well known and perfectly taken into account.

Radiance Temperature Relationship

The satellite digital numbers (DN) are converted into radiances and then into brightness temperature values. For this, calibration is performed and then the spectral radiance I_λ is converted into the brightness temperature T_s, which is the temperature of a blackbody emitting the same radiance than the observed one, by use of Planck's law:

$$B_\lambda(T) = 2hc^2 / \lambda^5[\exp(hc / \lambda KT) - 1] , \qquad (3)$$

where h is Planck's constant, c is the velocity of light, and K is Boltzmann's constant. Equation (3) is valid for the ideal monochromatic case, and should be adapted for sensors having a limited spectral response. Consequently, for any temperature retrieval, it is necessary to take into account the sensor-normalized response function.

This first section presented the main concepts necessary to compute a surface brightness temperature provided that the sensor, the atmosphere, and the surface emissivity are perfectly

known. After a brief description of the AVHRR, we are going to see what is actually "known" (sensor), what has to be assessed (atmosphere and emissivity), and how to do it.

THE AVHRR

The NOAA meteorological polar orbiters are sun-synchronous satellites whose altitude is nominally 825 km. They carry a scanning radiometer: the AVHRR. We will consider hereafter the case of the AVHRR/2 on board the odd-number NOAA satellites, since only this version of the AVHRR has two different thermal infrared channels. The AVHRR/2 has five channels in the short wave (red and near infrared), mid-infrared, and thermal infrared. The AVHRR/2 field of view is of $\pm 55°$, which enables the system to view almost any point of the Earth's surface twice a day (ascending and descending orbits). Nominally (i.e., without considering the drift of the satellite), overpass time is around 2 p.m. local solar time. Even though a given point of the surface is viewed every day, it must be noted that it will be viewed with different viewing angles, the viewing conditions being approximately repeated every 9 days.

The AVHRR/2 spectral bands are: 0.58–0.68 μm, 0.725–1.1 μm, 3.55–3.93 μm, 10.30–11.30 μm, and 11.50–12.50 μm. Subsatellite resolution is of the order of 1.1 km. Algorithms using the 3.7 μm channel will not be discussed here since they can only be efficiently used at night due to the reflected solar signal, and since this channel is severely affected by noise.

The first two bands are in the short wave part of the spectrum and are widely used to derive the normalized difference vegetation index (NDVI), which is the ratio of the difference to the sum of the reflectances ρ_1 and ρ_2:

$$\mathrm{NDVI} = \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1}. \tag{4}$$

It has been shown (e.g., Tucker and Sellers, 1986) that this ratio can be used to monitor biophysical properties of vegetation such as the leaf area index (LAI)—which is the total area of the leaves per unit area—and photosynthetic capacity. The NDVI can also be used to quantify the vegetation cover when the LAI does not exceed 3 due to a saturation effect.

Thermal band calibration is rather straightforward. The sensor views thermistances and deep space, which gives the calibration curve (Kidwell, 1986). Nonlinearities should be taken into account (Brown, 1985). The procedure is simple and reliable, even though some questions were recently raised concerning the "hot target" blackbodies. Consequently, deriving brightness temperatures at the top of the atmosphere is relatively simple and reliable (Kerr, 1986). The problem we will study now is the atmospheric correction procedure. The errors induced by the atmospheric contribution will be especially large for hot surfaces with humid atmospheres.

ATMOSPHERIC CORRECTIONS

The first method one can think of to perform atmospheric corrections is to use a radiative transfer model coupled with a characterization of the atmosphere. The characterization can be made from "standard values" such as the climatology means, but this characterization is bound to introduce errors due to spatial and temporal variability of the atmosphere. Another possibility is to use the output of meteorological forecasting models, but the method is cumbersome and not very accurate over some parts of the world where the ground network is not very dense. Using indirect methods has also been suggested, such as the use of a reference target (typically a large water body) of known temperature (which is another challenge) to assess atmospheric contribution, assuming that the atmosphere characteristics will not change spatially, which, obviously, is not the general case. Moreover, this method relies on only one measurement, which is a "cold" reference in the case of a water body, when at least two are necessary.

It is thus necessary to use more accurate characterizations of the atmosphere. Several methods have been thought of to assess the pressure, temperature, humidity (PTU) profile of the atmosphere. The most evident being to use radiosoundings. The PTU profile can then be used as an input to a radiative transfer code such as the "4A" (Scott and Chedin, 1981), LOWTRAN (Kneizys et al., 1983), or WINDOW (Price, 1983). This method can give very satisfactory results, provided that the radiosoundings are synchronous and colocated with the satellite measurement.

Otherwise, large errors can be introduced [up to 10 K as shown by Cooper and Asrar (1989)]. Moreover, the use of radiosoundings is hampered by the insufficient density of the network in some areas (3 for the whole Sudanian Sahel), by the timing (usually 12:00 local time), which is not the satellite overpass time, by the poor representativity in some cases (near the coast in arid areas for example), and by the difficulty in having the data in a timely fashion or in digital form. So it seems that ground-based radiosoundings do not really fit our needs. An alternative could be to use atmospheric profiles derived from satellite measurements (Susskind et al., 1984; Chedin et al., 1985), but in this case the inversion algorithms are time-consuming and very complex. Moreover, the existing sounders do not have the capacity to probe near the surface where most of the atmospheric water vapor is located. Such methods are thus not yet relevant, but they will have to be looked at when we enter the Earth Observing System (EOS) era since the NASA polar platform will carry a "super AVHRR"—MODIS-N—and a "super" TOVS—AIRS. The simultaneous use of both sensors should allow us to derive accurate surface brightness temperatures.

In the meantime, it seems necessary to look for alternative methods which, even though less accurate theoretically, have the advantage of being suitable for the whole globe. We have investigated the differential absorption method.

The Differential Absorption Method

Background

When two channels, or more, corresponding to different atmospheric transmissions, are available, it is possible to use the differential absorption to estimate the atmospheric contribution to the signal. This method was first suggested by Anding and Kauth (1970) and put in its now "classical" form by Prabhakara et al. (1974). It has been since adapted and tested with AVHRR data, mainly over sea surfaces, with success (Njoku, 1985). Its general name is the split window technique (SWT). The SWT has been tested over land surfaces as well, but with varying degrees of success (Price, 1984; Lagouarde and Kerr, 1985; Cooper and Asrar, 1989). The SWT relies on the different absorption characteristics of the atmosphere within two different but close wavelengths. The algorithm consists simply of a linear combination of the thermal channels which gives a surface temperature corrected for atmospheric contribution. For the AVHRR/2 the equation is

$$T_s = a_0 + a_1 T_{10.8} + a_2 T_{11.9}\,, \tag{5}$$

with

$$a_1 + a_2 = 1\,, \tag{6}$$

where $T_{10.8}$ and $T_{11.9}$ are the brightness temperatures at the top of the atmosphere in the two infrared bands. The a_i coefficients are estimated with various methods, depending on the authors. The SWT is now used operationally over oceans with a claimed accuracy of 0.2 K.

It must be noted that an alternative method, consisting in using different view angles, can be used. It can rely on measurements made by two different satellites (Becker, 1982; Chedin et al., 1982), or the same satellite provided that it can view along track with two different angles. The Along Track Scanning Radiometer (ATSR) on board ERS-1 will satisfy this dual viewing angle and differential absorption technique simultaneously.

Even though the SWT works satisfactorily over sea surfaces, when used over land surfaces, the errors can reach 6 K (Lagouarde and Kerr, 1985). This is mainly due to the fact that the assumptions made for the SWT over sea surfaces are not applicable over land surfaces. We are now going to study the SWT assumptions and describe why they are not always valid for land surfaces. We will then describe how the SWT could be adapted and used over such surfaces.

Analysis of the SWT

The SWT has been developed for sea surfaces. It is a simplified way to take into account atmospheric effects and thus relies on a number of assumptions such as: a) the surface is lambertian; b) the surface temperature is close to the temperature in the lower layers of the atmosphere, the latter varying slowly (Plank's law linearization); c) the surface temperature does not exceed 305 K; d) absorption in the atmosphere is small and occurs essentially in the lower layers; e) the surface emissivity is very stable spatially and close to unity; f) the emissivities $\epsilon_{10.8}$ and $\epsilon_{11.9}$ are almost identical and $\epsilon_{10.8} > \epsilon_{11.9}$. It is obvious that these

conditions are not usually met over land surfaces; hence the problems encountered when using the SWT over land surfaces. We felt, nevertheless, that provided we accept a somewhat reduced accuracy, the SWT could be adapted to land surface temperature retrieval.

It should also be kept in mind that, in the specific case of the AVHRR, several limitations linked to the instrument itself are to be considered: a) the sensor saturates for temperatures higher than 320 K; b) the ascending node time drifts constantly; c) due to its large field of view, the sensor views simultaneously points whose local solar time are quite different (almost 2 h from one end of the scan to the other); d) for two successive overpass, the sensor views a given point with different angles and at different solar time. Thus, over heterogeneous areas, angular effects are bound to be present.

The perturbating effects on the SWT when used over land surfaces are mainly the following: The surface emissivity is *a priori* unknown and different from unity; spatial variability of the emissivity can be high; surface temperature has also a very high spatial variability even at scales smaller than the resolution of the AVHRR; a strong gradient between air and surface temperature may exist.

We are now going to analyze the influence of the emissivity on the split window algorithms. Land surface emissivity can be described as having two characteristics whose effects are negative:

The mean emissivity in the spectral band 10.3–12.5 μm is not equal to 0.99 but might range from 0.90 and 0.98 (Fuchs et al., 1967; Fuchs and Tanner, 1966; Buettner and Kern, 1965),

The spectral emissivity is not constant over the two AVHRR channels, and it has been shown by Becker (1987) that if we assumed that the spectral emissivities $\epsilon_{10.8}$ and $\epsilon_{11.9}$ were equal to 1 when they are actually different and different from 1, the error induced by such an assumption on the retrieved surface temperature using the SWT could be written as

$$\Delta T = 50[(1-\epsilon)/\epsilon] - 300[(\epsilon_{10.8} - \epsilon_{11.9})/\epsilon], \quad (7)$$

where

$$\epsilon = (\epsilon_{10.8} + \epsilon_{11.9})/2. \quad (8)$$

It can also be said that the difference $\Delta\epsilon = \epsilon_{10.8} - \epsilon_{11.9}$, when positive, reduces the errors since the second term compensates the first in Eq. (7). This case occurs with water and vegetation. Moreover, the closer to 1 is ϵ, the smaller will be the errors on T_s. In conclusion, the classical SWT will give good results over water, slightly less over fully vegetated areas, and poor results on dry bare soil.

It is thus necessary to know the two spectral emissivities to derive surface temperature, which gives us a system of two measurements for three unknowns. The problem is thus *a priori* not solvable. We will consequently see whether it is possible to establish a local SWT with coefficients a_i being functions of the surface. The coefficients a_i are either derived from exact knowledge of the emissivity or empirically.

DERIVING LOCAL SWT COEFFICIENTS FROM KNOWN EMISSIVITIES

In a recent paper Becker and Li (1990) proposed, after a linearization of the radiative transfer equations, a method to derive the coefficients a_i of the SWT. This method is only valid locally since emissivities have a high spatial variability. Consequently, as was suggested by Kerr and Lagouarde (1989), for land surfaces, a SWT can only work locally and has to be tailored to the surface characteristics. Becker and Li (1990) give an expression of the SWT, where the coefficients a_i are derived from the actual emissivity values. For the specific case of the AVHRR/2 on board the NOAA-9 satellite they obtain the analytical expression of the coefficients a_i of Eq. (5):

$$\begin{aligned} a_0 &= 1.274, \\ a_1 &= 3.63 + 2.068(1-\epsilon)/\epsilon + 18.924\Delta\epsilon/\epsilon^2, \\ a_2 &= -2.63 - 1.912(1-\epsilon)/\epsilon - 19.406\Delta\epsilon/\epsilon^2, \end{aligned} \quad (9)$$

where $\epsilon = (\epsilon_{10.8} + \epsilon_{11.9})/2$ and $\Delta\epsilon = (\epsilon_{10.8} - \epsilon_{11.9})/2$. This method has been tested in the framework of HAPEX MOBILHY (André et al., 1988) over one location in the southwest of France, for which the spectral emissivities had been estimated (Becker and Li, 1990). The results gained look good, but

the method has still yet to be validated over other types of surfaces. The main results from this study are that the coefficients a_i are independent of the atmosphere and can be derived directly from the knowledge of the spectral emissivities. Values of surface emissivities can be found in the literature, even though they are scarce (Fuchs et al., 1967; Fuchs and Tanner, 1966; Buettner and Kern, 1965; Takashima and Matsuda, 1988; Elvidge, 1988; Eastes, 1989), but we do not have access yet to global data sets of emissivity with sufficient spatial resolution. To apply such a method routinely, global atlases of emissivities are required. In the EOS time frame, the ASTER sensor might prove to be a very useful tool to built such atlases, which do not yet exist.

SEMIEMPIRICAL SWT FOR LAND SURFACES

Kerr and Lagouarde (1989) proposed a method similar to that of Becker and Li (1990) except that it tries to avoid the necessity of having access to exact values of the spectral emissivity. It must be said that the method was primarily designed for arid and semiarid areas, where the vegetation can be very sparse and the surface temperature high. Consequently, the method is somewhat similar to the Becker and Li (1990) method with the addition of two improvements: The method takes into account the canopy fraction and tries to cope with angular (shadowing) effects (Kimes and Kirchner, 1983).

Method Used

Emissivities

For bare soil, if the emissivities are known for each pixel, the Becker and Li method can be used to compute the SWT coefficients a_i. Such is not generally the case, and when unknown, the coefficients are derived empirically for bare soil from ground measurements. Not having any emissivity data for our test sites, we derived the coefficients a_i empirically. For this, half the data set collected over one test site was used to derive the SWT coefficients. The coefficients obtained were then used for both sites. Considering the resolution of the AVHRR on the one hand and the resolution of ground measurements on the other hand, such measurements are difficult to perform. It requires having large areas of homogeneous bare soil. Moreover, the spectral bands of ground thermal radiometer are most often "broad band," that is, 8–14 μm and comparison with AVHRR measurements are difficult.

For vegetation covered areas, we can consider that the emissivities are those of water and apply directly sea surface temperature coefficients. Better still, actual values of emissivities obtained from the literature can be used when available.

Angular Effects

Two types of angular effects can affect the measurements:

a. Angular effects linked to atmosphere: These are negligible on the SWT coefficients (Ottlé and Vidal-Madjar, 1992).
b. Angular effects linked to the surface: For vegetation, angular effects due to shadowing can be quite significant (Kimes and Kirchner, 1983). As for the AVHRR, from one acquisition to the other, the viewing and solar angle can be significantly different, and it is necessary to take this effect into consideration. Not having any tractable analytical model for the angular effects available, we had to use a very empirical and approximate formulation to correct for angular effects. From ground measurements made at various solar and viewing angles, we derived a function which best fitted the measurements:

$$\gamma = \alpha[0.74 + 0.19\delta_\theta + 0.07\delta_\theta^2] \qquad (10)$$

with $\delta_\theta = \theta_s - \theta_v$, where θ_s is the solar zenith angle, θ_v is the viewing zenith angle, and α is a factor linked to the difference in azimuth between the solar and view angle. In our case, α does not vary much and $\alpha = 1.03$.

The corrected temperatures $T_{\lambda c}$ are then given by

$$T_{\lambda c} = T_\lambda / \gamma . \qquad (11)$$

The function given in Eq. (10) is very emipirical, valid only for $\delta_\theta < \pi / 3$, and is only representative of the surface over which the measurements were made. Consequently, we will not describe the angular correction in this paper and give only results obtained without performing any angular

corrections. Simply, just to show what the improvement is brought by an even very coarse correction, the curve with angular corrections performed is given in Figure 1. More work is cleary required in this area of angular effects in the thermal infrared for natural vegetation.

Another point is worth noting: The function given in Eq. (10) was established from ground measurements. It is not clear yet how valid such a relationship over heterogeneous surfaces and for an AVHRR pixel is. Most probably, due to averaging effects, angular effects will be significantly reduced, making Eq. (10) unnecessary.

Fractional Vegetation Cover

In arid and semiarid areas, vegetation cover can be sparse and may also evolve rapidly with time. It is thus necessary to take this factor into account since we will almost never have pure pixels of bare soil or vegetation. In order to do this, we considered that in arid and semiarid areas the leaf area index never exceeds 3 and that consequently we could express the fractional cover as a function of the normalized difference vegetation index (NDVI).

Algorithm

For AVHRR data, we compute for each pixel the NDVI and the top of atmosphere temperature in the two thermal bands. It is then necessary assess the bare soil SWT coefficients a_{bsi} for each pixel. This can be achieved through an *a priori* knowledge of the soils (soil maps) and the knowledge of the spectral emissivities of each soil type. When no soil maps are available at the appropriate scale, it is possible to use image classification techniques to get an estimate of the soil types, provided that some ground data are available. Once we know the soil type and the corresponding emissivities for each pixel, we can derive the bare soil SWT coefficients a_{bsi} using an expression such as the one given in Eqs. (9). When the emissivities are unknown, we can use empirically derived a_{bsi} for each estimated soil type.

Once the a_{bsi} coefficients are known, it is possible to compute, using the SWT, two surface temperatures: T_{bs} for bare soil with the derived coefficients a_{bsi} and T_v for vegetation with the coefficients a_{vi} computed for vegetation emissivities.

$$T_{bs} = a_{bs0} + a_{bs1}T_{10.8} + a_{bs2}T_{11.9}\,, \tag{12}$$

$$T_v = a_{v0} + a_{v1}T_{10.8} + a_{v2}T_{11.9}\,. \tag{13}$$

When angular corrections are performed, they are applied to the brightness temperatures $T_{10.8}$ and $T_{11.9}$. A fractional cover coefficient C is then derived from the NDVI values with the expression

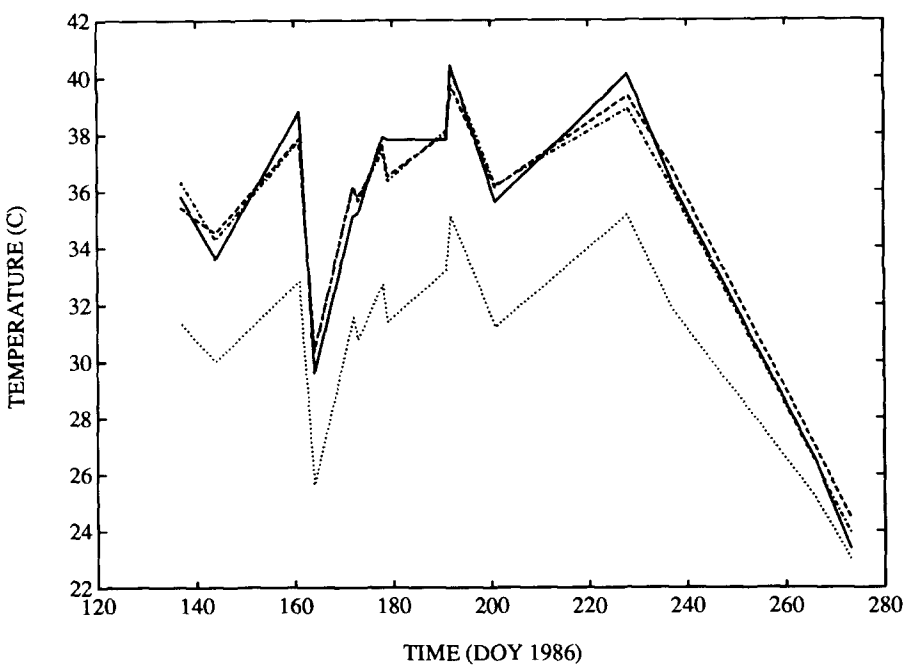

Figure 1. Avignon Caumont test site: Temporal evolution of the measured surface temperature (——) and the surface temperature estimated from satellite data with a) the sea surface temperature SWT (· · ·), b) the algorithm described in the article (- · - · -), and c) the same algorithm with angular corrections performed (- - -).

$$C = \frac{(\mathrm{NDVI} - \mathrm{NDVI_{bs}})}{(\mathrm{NDVI_v} - \mathrm{NDVI_{bs}})}, \quad (14)$$

where $\mathrm{NDVI_{bs}}$ is the minimum value of the NDVI for bare soil over the area of interest (typically end of the dry season in the Sahel) and $\mathrm{NDVI_v}$ correspond to the highest NDVI you could expect for a fully vegetated pixel (typically end of the rain season in the Sahel). This formulation is dependent upon the level of atmospheric corrections performed, as explained in the data processing chapter.

The fractional vegetation cover is computed on a pixel per pixel basis and for every date since it varies with time and space. The expression given in Eq. (14) is very coarse and rather arbitrary. It could certainly be improved, though the increase of complexity might not induce a significant improvement of the global accuracy of the algorithm. The most promising improvement would be with the use of a new vegetation index, directly linked with vegetation cover and not sensitive to soil background or directional effects.

Finally a surface temperature (T_s) is estimated through a linear combination of T_{bs} and T_v:

$$T_s = CT_v + (1 - C)T_{bs} \quad (15)$$

We used a linear combination of the two temperatures (bare soil and vegetation) since it is simple to implement and since no theory background justifies a more elaborate formulation. The results gained with this formula indicates that this expression, even though simple, is adequate for our purpose.

RESULTS AND DISCUSSION

Data Used

Ground Data

Ground surface temperatures were measured with a radiothermometer (Heinman KT-17). The radiothermometer makes measurements in the range 8–14 μm. It was attached to a mast so that the field of view had a diameter ranging from 2 m to 4 m, depending on the heterogeneity of the surface [see Seguin et al. (1989) for a complete description of the setup]. The target aimed was as representative of the area as possible, and the shadow of the mast never fell within the field of view. Measurements were performed continuously with an integration time of 15 mn. The radiothermometer was sheltered from solar radiation by an aluminium shield. The radiothermometer is calibrated in the lab for various internal temperature values. During ground experiments, internal temperature of the radiothermometer is continuously monitored and used for calibration (Lagouarde et al., 1986).

Satellite Data

The data used consisted of NOAA-9 AVHRR/2 data preprocessed as described in Kerr (1986) with calibration performed using Holben (1990) values for the calibration of the short wave channels. The atmospheric corrections in the short wave bands were performed using the 5S algorithm (Tanre et al., 1990) with standard climatological mean values for ozone, visibility data for aerosols, and water vapor content derived from radiosonde data. For the French test site we had a significant number of AVHRR scenes (15 dates in 1986), but for the Niger test site we had only three cloud-free images in 1988.

Test Sites

The method presented has been tested and validated over two test sites. One is located in the southeast of France (Caumont near Avignon 43°54′N–4°51′E) (Kerr and Lagouarde, 1989) and the other in Niger in Danguey Gourou (13°59′N–2°E) near Niamey (Imbernon et al., 1990).

The Avignon Caumont is a small air field surrounded by dry natural grassland with patches of bare soil. However, by the end of the period of measurements, the surrounding fields have been irrigated leading to the growth of grass. So as to extract AVHRR data over our test site, the maximum values of temperature over a 2×2 pixel area surrounding the location of the ground station were selected for each date.

The Niger test site is representative of the vegetation cover in the Sahel. It is very heterogeneous at the subpixel scale, but homogeneous globally. The vegetation consisted mainly of millet, fallow, and tiger bush (degraded forest).

Data Processing

For each date and for both studies, the AVHRR data in the short wave and long wave channels

corresponding to the ground station locations were extracted. The short wave reflectances at the surface and the brightness temperatures at the top of the atmosphere were then computed. These were used to compute the NDVI and canopy fraction [Eq. (14)], the bare soil temperature [Eq. (12)], and the vegetation temperature [Eq. (13)].

The a_i coefficients used for both cases (France and Niger) were those derived empirically from the Caumont data set, given in Table 1. In order to derive the coefficients a_i, we took the measurements and satellite data between DOY 161 and DOY 201 (which corresponds to 60% of the data set) so as to determine the coefficients a_i giving the best fit. These coefficients were then used for the whole data set (both France and Niger) since the soil types are somewhat equivalent. The results are then combined with Eq. (15) to derive the surface temperature.

For the computation of C [Eq. (14)], we used for $NDVI_{bs}$ and $NDVI_v$ the values of 0.11 and 0.72, respectively. These values were obtained over a field of lush millet in Niamey by Begue (1991) with a field radiometer. Even though these values were derived in Niger, we also used them for the French test site, where vegetation and soil characteristics are somewhat similar to those of Niger.

It must be noted that these specific values of $NDVI_{bs}$ and $NDVI_v$ are only to be used with atmospherically corrected NDVI. As it is not always possible to get access to all the relevant atmospheric parameters on a routine basis, values for $NDVI_{bs}$ and $NDVI_v$ can be derived from satellite data for which only solar angle and Rayleigh scattering corrections are performed. Water vapor and ozone content can be extracted from climatological means. The errors induced will then be linked to the variability of the atmospheric characteristics around mean values, aerosol loading, and to how representative are $NDVI_{bs}$ and $NDVI_v$.

Results

Figure 1 shows the temporal evolution of the signal measured on the ground in Caumont and the surface temperature estimated with various SWT algorithms. It is evident that the sea surface SWT is not applicable for this case as it induces errors of the order of 5 K. The suggested algorithm fits rather well the measured data. The difference between measured and estimated surface temperature (Fig. 2) has a mean value of −0.25 K and a standard deviation of 0.9 K. The estimates are practically always within the ±1°C area. From Figure 2 it can be seen that all the estimates are within 1 K of the measured values. It can be noted in Figure 1 that, when angular corrections are performed, the mean difference between measurements and estimates drops to −0.1 K with a standard deviation of 0.8 K.

Table 1. Coefficients for the Split Window Technique Used in This Study

Surface Type	a_0	a_1	a_2
Bare soil	3.1	3.1	−2.1
Vegetation	−2.4	3.6	−2.6

Figure 3 shows the same data set as in Figures 1 and 2 but organized differently. It is a scatter plot of the measured and estimated surface temperatures. The points are well aligned with the 1:1 line over the whole range of measurements. The correlation coefficient is 0.992.

In Niger, we had much less data, and a statistical analysis is not significant. We, nevertheless, compared the ground measurements with the satellite estimates, and the results are given in Figure 4. The continuous curve corresponds to ground measurements (time step of 15 min) while the "asterisk" is the satellite estimate. Again a very good agreement is found.

Discussion

The method described in this article gives very satisfactory results. The accuracy is always better than 1.5 K, and we can safely generalize it to a global accuracy of 1.5 K. As a matter of fact, other land surfaces are bound to give better results since the arid land environment is the most difficult case for the SWT. Considering other existing techniques, the method has the advantage of being valid for arid and semiarid environments where the general assumptions of the SWT are most prone to be unsatisfied. It can, of course, be said that some steps of the algorithm rely on very empirical approaches. However, the method has

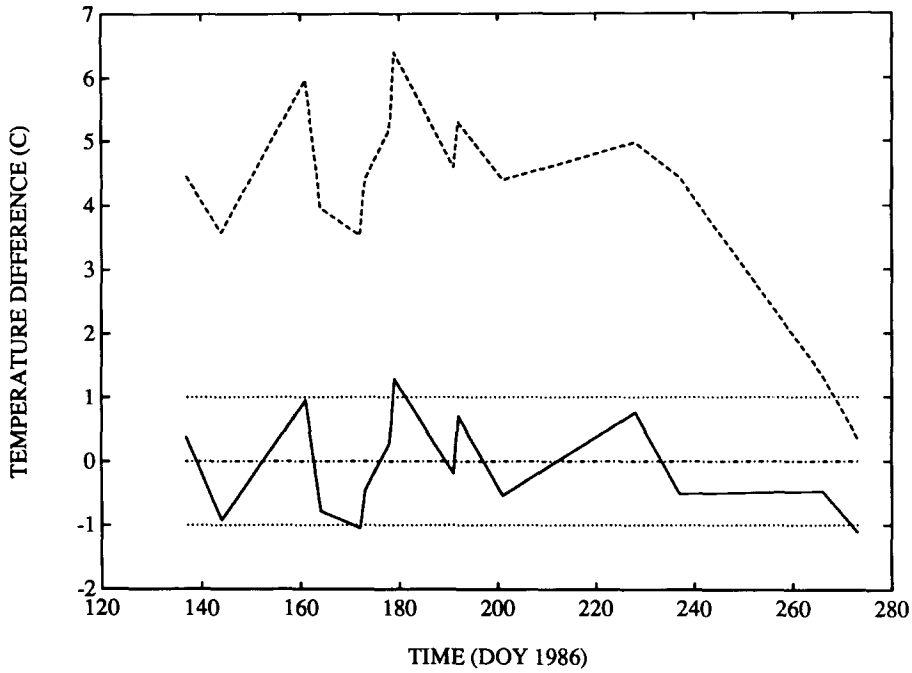

Figure 2. Avignon Caumont test site: Temporal evolution of the difference between the measured ground temperature and the temperature derived from the AVHRR data with a sea surface SWT (- - -), the algorithm described in this paper (——). The dotted lines correspond to the +1°C and −1°C values.

also some potential improvement through a better quantification of the canopy fraction and through a more accurate modeling of directional effects. It can also be improved by using a more physical determination of the SWT coefficients a_i, when the emissivities are known, by using the Becker and Li formulation.

In the Cooper and Asrar (1989) article, several algorithms were tested. The test area was mainly grassland, which is a surface type rather suitable for sea surface SWT algorithms. Nevertheless, they found that the radiosounding methods failed quite often, and that only one algorithm gave satisfactory results consistently, the McClain et al. (1983) SWT algorithm. It is very likely that those algorithms would have failed over a Sahelian type surface, as shown in Figure 1, where the SWT with sea surface coefficients gives a large error in the estimate of surface temperature. For instance, one can see in Figure 1 that when only using a classical SWT algorithm (i.e., for sea surfaces), the errors are large when the surface is bare, but almost disappear when vegetation grows (end of the data set, from DOY 250). It can also be noted in Figure 1 that variations in soil moisture do not affect the data. It is mainly due to the fact that in an arid or semiarid environment the soil's "skin" dries very quickly when the sun shines while satellite data is acquired shortly after noon and under clear sky conditions. Consequently, for bare soil, the very surface of the ground will be dry most of the time, which enforces the use of Eq. (15).

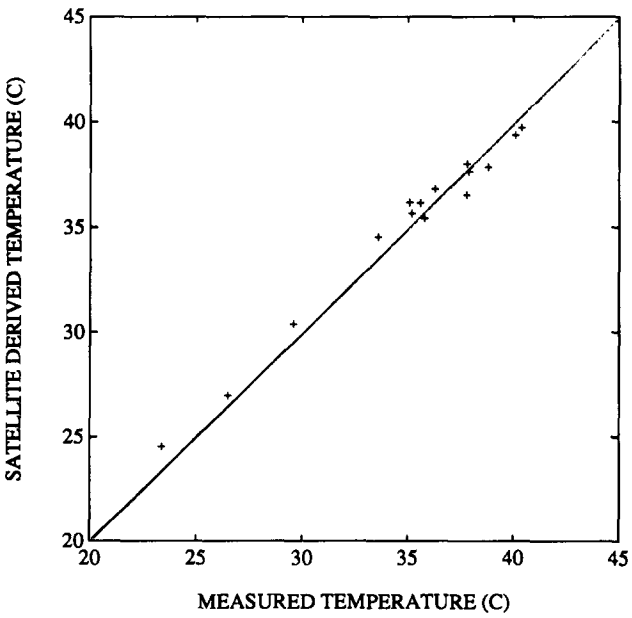

Figure 3. Avignon Caumont test site: Scatter plot of the measured and estimated surface temperatures.

We would also like to stress problems linked with validation. It is necessary to have ground and satellite measurements of surface temperature that are consistent. For this purpose, the sensors should have the same wave band and

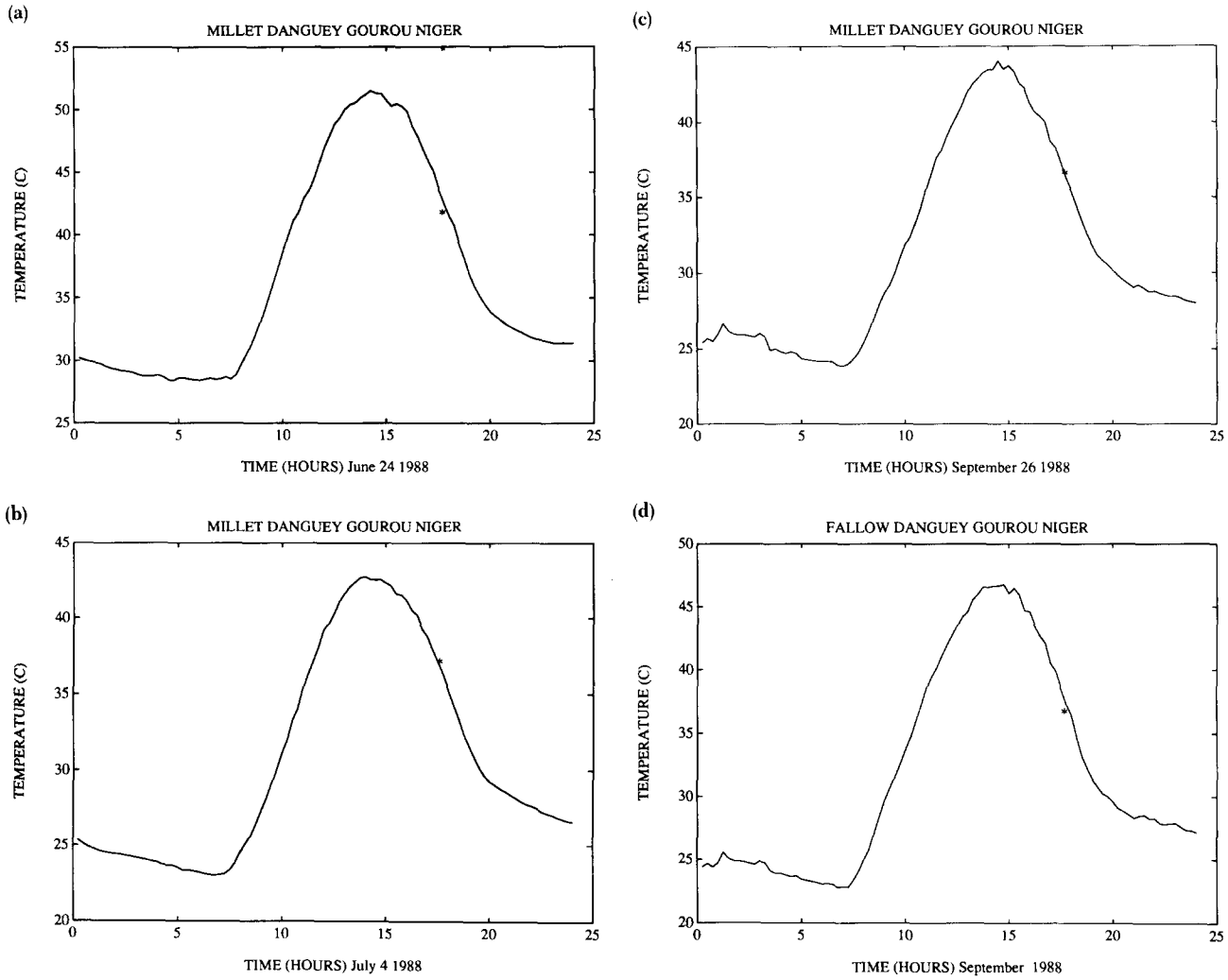

Figure 4. Niger test site: Estimated and measured surface temperature: (*) satellite-derived surface temperature; (——) ground measurements. a) 24 June 1988 millet field; b) 4 July 1998 millet field; c) 26 September 1988 millet field; d) 26 September 1988 fallow land.

should measure similar areas on the ground. This last point being almost impossible to satisfy practically, we will have to either find large homogeneous areas carefully sampled, or investigate in full depth the scaling up methods with use of measurements performed with varying resolutions (ground, aircraft, satellite). Several large scale experiments (FIFE, EFEDA-91, and HAPEX-SAHEL-92) should allow us to start investigating this very interesting problem in the thermal infrared. They will also enable us to validate further and improve algorithms such as the one described in this paper.

Obviously, the suggested algorithm has still to be validated for other areas of the globe. However, the fact that it works satisfactorily over arid and semiarid areas makes us confident that it will be suitable for most of the globe. We are currently investigating possible test sites for this purpose. Once fully validated, the algorithm will be rather simple to implement on a routine basis, provided that we have access to soil maps and values of $NDVI_{bs}$ and $NDVI_v$ on a regional scale.

CONCLUSION

Algorithms to retrieve surface temperature from space measurements are numerous. Over sea surfaces, the SWT has proved to be efficient and accurate in most of the cases. It is now used operationally. Over land surfaces, the classical

SWT does not give satisfactory results due to the spectral emissivity effects. We have thus derived a modified SWT, which is valid locally and taking into account the emissivities of each pixel. In order to work over areas of sparse vegetation, the algorithm takes into account a canopy fraction derived from the NDVI. Finally, it is also possible to take into account angular effects on the signal.

The suggested method gives very satisfactory results (1.5 K accuracy) and can be easily implemented in an operational or semioperational processing scheme. It can also be upgraded to take into account any significant progress in canopy fraction determination and angular response modeling. It should also be easy to improve it with the geocoded information system (GIS) of soil surface types or emissivity for the determination of the SWT coefficients for bare soil (vegetation is easier and varies continuously with time).

Such an algorithm can be coupled with a SST SWT algorithm to derive surface temperature globally, and can be used routinely without the need for auxiliary data (atmospheric profiles) other than data that do not vary with time (surface type or emissivity), since the evolution of vegetation cover is taken into account by the algorithm itself.

The Niger data set was acquired during an experiment funded by the European Community Commission, in the framework of the program: "Caractérisation par les techniques de la télédétection de la dynamique de la désertification à la périphérie sud du Sahara." The authors are indebted to the reviewers for comments and suggestions that have improved the manuscript.

REFERENCES

Anding, D., and Kauth, R. (1970), Estimation of sea surface temperature from space, *Remote Sens. Environ.* 1:217–220.

Andre, J. C., et al. (1988), HAPEX-MOBILHY: first results from the Special Observing Period, *Ann. Geophys.* 6:447–492.

Becker, F. (1982), Absolute sea surface temperature measurements by remote sensing and atmospheric corrections differential radiometry, in *Processing in Marine Remote Sensing*, University of South Carolina Press, Columbia, pp. 151–174.

Becker, F. (1987), The impact of spectral emissivity on the measurement of land surface temperature from a satellite, *Int. J. Remote Sens.* 8(10):1509–1522.

Becker, F., and Li, Z. L. (1990), Towards a local split window method over land surfaces, *Int. J. Remote Sens.* 11(3):369–393.

Begue, A. (1991), Estimation de la production primaire en zone sahélienne à partir de données radiométriques. Cas d'un couvert discontinu: le mil, Ph.D. Thesis, Université de Paris VII, UFR de Physique, 1991, 139 pp.

Brown, O. B., et al. (1985), Calibration of AVHRR infrared observations, *J. Geophys. Res.* 90:11667.

Buettner, K. J. K., and Kern, C. D. (1965), The determination of infra-red emissivities of terrestrial surfaces, *J. Geophys. Res.* 70:1329–1337.

Chandrasekhar, S. (1960), *Radiative Transfer*, Dover, New York.

Chedin, A., Scott, N. A., and Berroir, A. (1982), A single channel double viewing angle method for sea surface temperature determination from coincident METEOSAT and TIROS-N radiometric measurements. *J. Clim. Appl. Meteorol.* 21:613–618.

Chedin, A., Scott, N. A., Wahiche, C., and Moulinier, P. (1985), The Improved Initialization Inversion method: a high resolution for temperature retrievals from the TIROS-N series, *J. Clim. Appl. Meteorol.* 24:124–143.

Cooper, D., and Asrar, G. (1989), Evaluating atmospheric correction models for retrieving surface temperature from the AVHRR over a tall grass prairie, *Remote Sens. Environ.* 27:93–102.

Eastes, J. W. (1989), Spectral properties of halite rich mineral mixtures: implications for middle infra-red remote sensing of highly saline environments, *Remote Sens. Environ.* 27: 289–304.

Elvidge, C. D. (1988), Thermal infra-red reflectance of dry plant materials: 2.5–20 μm, *Remote Sens. Environ.* 26: 265–285.

Fuchs, M., and Tanner, C. B. (1966), Infra-red thermometry of vegetation, *Agron. J.* 58:597–601.

Fuchs, M., Kanemasu, E. T., and Kerr, J. P. (1966), Effect of viewing angle on canopy temperature measurements with infra-red thermometer, *Agron. J.* 59:494–496.

Holben, B. N., Kaufman, Y. J., and Kendall, J. D. (1990), NOAA-11 AVHRR visible and near-IR inflight calibration, *Int. J. Remote Sens.* 11:1511–1519.

Imbernon, J., et al. (1990), Suivi Agroclimatique des cultures en zone sahélienne à partir des données NOAA de Mas Palomas, Final Report, CEE, November 1990, 166 pp.

Kerr, Y. H. (1986), NOAA / AVHRR: a unique tool for Earth's resources monitoring, in *1st Australian AVHRR Conference*, Perth, Australia, pp. 127–137.

Kerr, Y. H., and Lagouarde, J. P. (1989), On the derivation of land surface temperature from AVHRR data, *EUM-P06*, pp. 157–160, Eumetsat, Rothenburg, FR Germany.

Kidwell, K. B. (1986), *AVHRR User's Guide*, U.S. Dept. of Commerce, NOAA / NESDIS, Washington, DC, 128 pp.

Kimes, D. S., and Kirchner, J. A. (1983), Directional radio-

metric measurements of row crop temperatures, *Int. J. Remote Sens.* 4:299–311.

Kneizys, F. X., Shettle, E. P., Gallery, W. O., et al. (1983), Atmospheric transmittance / radiance: computer code Lowtran 6, Technical Report AFGL-TR-83-0187, Optical Physics Division, U.S. Air Force Geophysics Laboratory, Hanscom Air Force Base, MA.

Lagouarde, J. P., and Kerr, Y. H. (1985), On the calibration of the thermal NOAA / AVHRR sensor, *ESA SP* 248:325.

Lagouarde, J. P., Kerr, Y. H., Assad, E., and Cognet, M. (1986), Résultats d'une campagne de mesures micrometéorologiques au Sénégal pendant l'hivernage 1985, INRA Internal Note M / 86 / 7, Montfavet, France, 27 pp.

McClain, E. P., Pichel, W. G., Walton, C. C., Ahmad, Z., and Sutton, J. (1983), Multichannel improvements to satellite derived global sea surface temperatures, *Adv. Space Res.* 2:43–47.

McClatchey, R. A., Fenn, R. W., Selby, J. E. A., Volz, F. E., and Garing, J. S. (1971), Optical properties of the atmosphere (revised), Environmental Research Paper N. 311.910.8, Air Force Cambridge Research Laboratories, 24 pp.

Njoku, E. G. (1985), Satellite derived sea surface temperature workshop comparisons, *Bull. Am. Meteorol. Soc.* 66: 274–281.

Ottlé, C., and Vidal-Madjar, D. (1992), Estimation of land surface temperature with NOAA9 data, *Remote Sens. Environ.* 40:27–41.

Prabhakara, C., Dalu, G., and Kunde, V. G. (1974), Estimation of sea surface temperature from remote sensing in the 11 and 13 μm window region, *J. Geophys. Res.* 79: 5039–5044.

Price, J. C. (1983), Estimating surface temperature from satellite thermal infra-red data—a simple formulation for the atmospheric effect, *Remote Sens. Environ.* 13:353–361.

Price, J. C. (1984), Land surface temperature measurements from the split window channels of the NOAA 7 Advanced Very High Resolution Radiometer, *J. Geophys. Res.* 89(D5):7231–7237.

Scott, N. A., and Chedin, A. (1981), A fast line by line method for atmospheric absorption computations: the automatized atmospheric absorption atlas, *J. Clim. Appl. Meteorol.* 20:801–812.

Seguin, B., Assad, E., Freteaud, J. P., Imbernon, J., Kerr, Y. H., and Lagouarde, J. P. (1989), Use of meteorological satellites for water balance monitoring in Sahelian regions, *Int. J. Remote Sens.* 10:1101–1117.

Susskind, J., Rosenfield, J. Renter, D., and Chahine, M. T. (1984), Remote sensing of weather and climate parameters from HIRS2 / MSU on TIROS / N., *J. Geophys. Res.* 89:4677–4697.

Takashima, T., and Matsuda, K. (1988), Averaged emissivities of quartz and Sahara dust powders in the infra-red region, *Remote Sens. Environ.* 26:301–302.

Tanré, D., Deroo, C., Duhaut, P., et al. (1990), Description of a computer code to simulate the satellite signal in the solar spectrum: the 5S code, *Int. J. Remote Sens.* 11:659–668.

Tucker, C. J., and Sellers, P. J. (1986), Satellite remote sensing of primary production, *Int. J. Remote Sens.* 7: 1395–1416.